

A Modified Parameterized Gaussian Error Linear Unit

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1 Background

Definition 1.1. *The normal distribution with parameter values $\mu = 0$ and $\sigma = 1$ is called the standard normal distribution. A random variable having a standard normal distribution is called a standard normal random variable and will be denoted by Z . The probability density function of Z is*

$$f(z) = \frac{1}{\sqrt{2\pi}} e^{-z^2/2}$$

where $z \in \mathbb{R}$. The cumulative distribution function of Z is

$$\mathbb{P}(Z \leq z) = \int_{-\infty}^z f(t) dt$$

which we denote by $\Phi(z)$.

1.1 Gaussian Error Linear Unit (GELU) Formulation

We first observe the similarities between the Rectified Linear Unit (ReLU) [3] and the Heaviside function.

$$\text{ReLU}(x) = \max(0, x) = \begin{cases} x, & \text{if } x > 0 \\ 0, & \text{if } x \leq 0 \end{cases} \quad \text{and} \quad H(x) = \begin{cases} 1, & \text{if } x > 0 \\ 0, & \text{if } x \leq 0 \end{cases}$$

As seen in their definitions, the Rectified Linear Unit (ReLU) can be explicitly expressed as the input x multiplied by the Heaviside step function $H(x)$, yielding $\text{ReLU}(x) = x \cdot H(x)$. This acts as a hard, deterministic gating mechanism. In contrast, regularization techniques like Dropout apply a stochastic gate, scaling the input by a random mask $m \sim \text{Bernoulli}(p)$. The stochastic dropout operation $\nu(x)$ is formulated as:

$$\nu(x) = m \cdot x = \begin{cases} x, & \text{with probability } p \\ 0, & \text{with probability } 1 - p \end{cases}$$

The expected value of this stochastic transformation is $\mathbb{E}[\nu(x)] = x \cdot p + 0 \cdot (1 - p) = x \cdot p$. Motivated by this, Hendrycks and Gimpel [2] sought to merge these paradigms by applying a zero-one mask that is stochastically determined yet strictly dependent upon the input. They defined this stochastic activation as:

$$f(x) = m \cdot x$$

where $m \sim \text{Bernoulli}(\Phi(x))$. Rather than dropping out neurons uniformly with a constant probability p , the Gaussian Error Linear Unit (GELU) defines the probability of preserving the input as a function of the input itself, specifically utilizing the standard normal cumulative distribution function. By replacing p with $\Phi(x)$, GELU is mathematically formulated as the expected value of this input-dependent stochastic process:

$$\text{GELU}(x) = \mathbb{E}[m \cdot x] = x\Phi(x).$$

Here, $\Phi(x)$ is the cumulative distribution function of the standard normal distribution, defined as:

$$\Phi(x) = \mathbb{P}(X \leq x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^x e^{-t^2/2} dt.$$

This specific distribution is used because neuron inputs naturally tend to follow a normal distribution, particularly when the network applies Batch Normalization. To evaluate the cumulative distribution function, we can split $\Phi(x)$ into two distinct intervals at zero:

$$\Phi(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^x e^{-t^2/2} dt = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^0 e^{-t^2/2} dt + \frac{1}{\sqrt{2\pi}} \int_0^x e^{-t^2/2} dt$$

The first term represents the left half of the probability distribution. To understand its exact value, we must first determine the total area under the foundational Gaussian curve:

$$\int_{-\infty}^{\infty} e^{-x^2} dx$$

We start by squaring the integral and introducing two independent dummy variables, x_1 and x_2 . Fubini's Theorem allows us to combine the product of two one-dimensional integrals into a single double integral over the entire Cartesian plane:

$$\begin{aligned} \left(\int_{-\infty}^{\infty} e^{-x^2} dx \right)^2 &= \left(\int_{-\infty}^{\infty} e^{-x_1^2} dx_1 \right) \left(\int_{-\infty}^{\infty} e^{-x_2^2} dx_2 \right) \\ &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} e^{-(x_1^2+x_2^2)} dx_1 dx_2 \end{aligned}$$

Next, we apply a change of variables to polar coordinates. Recall that for a continuous function on a polar region,

$$\iint_R f(x, y) dA = \int_{\alpha}^{\beta} \int_a^b f(r \cos \theta, r \sin \theta) r dr d\theta$$

Applying the substitutions $x_1^2 + x_2^2 = r^2$ and $dx_1 dx_2 = r dr d\theta$, and expanding the bounds to cover the entire infinite plane (r from 0 to ∞ and θ from 0 to 2π), we obtain:

$$\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} e^{-(x_1^2+x_2^2)} dx_1 dx_2 = \int_0^{2\pi} \int_0^{\infty} e^{-r^2} r dr d\theta$$

We evaluate the inner integral with respect to r using a standard u -substitution (where $u = -r^2$ and $du = -2r dr$):

$$\begin{aligned} \int_0^{2\pi} \int_0^{\infty} e^{-r^2} r dr d\theta &= \int_0^{2\pi} \left[-\frac{1}{2} e^{-r^2} \right]_{r=0}^{r=\infty} d\theta \\ &= \int_0^{2\pi} \left[0 - \left(-\frac{1}{2} \right) \right] d\theta \\ &= \int_0^{2\pi} \frac{1}{2} d\theta \end{aligned}$$

Evaluating the final outer integral with respect to θ :

$$\int_0^{2\pi} \frac{1}{2} d\theta = \left[\frac{1}{2} \theta \right]_0^{2\pi} = \pi$$

Thus, the square of our original integral equals π . Taking the square root of both sides yields the foundational result:

$$\left(\int_{-\infty}^{\infty} e^{-x^2} dx \right)^2 = \pi \implies \int_{-\infty}^{\infty} e^{-x^2} dx = \sqrt{\pi}$$

Now, we apply this result to find the total area under our unscaled standard normal curve:

$$\text{Total Area} = \int_{-\infty}^{\infty} e^{-z^2/2} dz$$

We evaluate this using another u -substitution. Let $x = \frac{z}{\sqrt{2}}$, which implies $z = x\sqrt{2}$ and $dz = \sqrt{2} dx$. Substituting this into the integral and keeping the infinite bounds, we get:

$$\int_{-\infty}^{\infty} e^{-x^2} (\sqrt{2} dx) = \sqrt{2} \int_{-\infty}^{\infty} e^{-x^2} dx$$

For a function to qualify as a valid probability density function, the total area under the curve must integrate to exactly 1. Dividing the raw kernel by its total area establishes our normalization factor:

$$\int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-z^2/2} dz = \frac{1}{\sqrt{2\pi}} \cdot \sqrt{2\pi} = 1$$

Because the curve is perfectly symmetric around zero and the total normalized area is exactly 1, the integral of the left half (from $-\infty$ to 0) evaluates to precisely to:

$$\frac{1}{\sqrt{2\pi}} \int_{-\infty}^0 e^{-z^2/2} dz = \frac{1}{2}$$

This allows us to resolve the first term of our split $\Phi(x)$ function, simplifying it to:

$$\Phi(x) = \frac{1}{2} + \frac{1}{\sqrt{2\pi}} \int_0^x e^{-t^2/2} dt$$

While the integral for $\Phi(x)$ perfectly describes the theory, practical implementations in deep learning frameworks (such as PyTorch [4]) rely on the Error Function:

$$\text{erf}(z) = \frac{2}{\sqrt{\pi}} \int_0^z e^{-u^2} du$$

Let $u = \frac{t}{\sqrt{2}}$. Then, $t = u\sqrt{2}$ and $dt = \sqrt{2} du$. For the limits of integration, we have:

If $t = 0$ then $u = 0$

If $t = x$ then $u = \frac{x}{\sqrt{2}}$

Substituting this into $\frac{1}{\sqrt{2\pi}} \int_0^x e^{-t^2/2} dt$, we obtain:

$$\frac{1}{\sqrt{2\pi}} \int_0^x e^{-t^2/2} dt = \frac{1}{\sqrt{2\pi}} \int_0^{x/\sqrt{2}} e^{-u^2} (\sqrt{2} du) = \frac{1}{\sqrt{\pi}} \int_0^{x/\sqrt{2}} e^{-u^2} du \quad (1)$$

And we can express (1) as:

$$\frac{1}{\sqrt{\pi}} \int_0^{x/\sqrt{2}} e^{-u^2} du = \frac{1}{2} \left[\frac{2}{\sqrt{\pi}} \int_0^{x/\sqrt{2}} e^{-u^2} du \right] = \frac{1}{2} \text{erf} \left(\frac{x}{\sqrt{2}} \right)$$

Finally, we obtain:

$$\text{GELU}(x) = x\Phi(x) = \frac{x}{2} \left[1 + \text{erf} \left(\frac{x}{\sqrt{2}} \right) \right]$$

The error function is a standard mathematical tool specifically built to calculate areas under Gaussian curves, and it is highly optimized at the hardware level. By rewriting our cumulative distribution function $\Phi(x)$ in terms of the error function, we can compute the exact GELU efficiently in software.

2 Proposed Activation Function

2.1 Formulation

The standard Gaussian Error Linear Unit (GELU) relies on the fixed variance of the standard normal distribution $\mathcal{N}(0, 1)$, which limits the network's ability to adapt to varying data distributions across layers. To address this, we propose a parameterized extension by introducing a learnable scaling parameter $\lambda \geq 1$.

To strictly enforce this lower bound during unconstrained gradient-based optimization, we introduce a parameter $s \in \mathbb{R}$ and map it through the Softplus function:

$$\lambda = 1 + \ln(1 + e^s)$$

Using this constrained parameter, we formulate our proposed activation function by scaling the input within the cumulative distribution function:

$$f(x) = x\Phi(\lambda x) = \frac{x}{2} \left[1 + \operatorname{erf} \left(\frac{\lambda x}{\sqrt{2}} \right) \right]$$

2.2 Gradient Formulation

To ensure the proposed activation function is compatible with gradient-based optimization during back-propagation. To rigorously evaluate the derivative of the cumulative distribution function, we rely on the foundational principles of calculus.

Theorem 2.1 (The First Fundamental Theorem of Calculus). *If f is continuous on $[a, b]$, then the function g defined by*

$$g(x) = \int_a^x f(t)dt \quad a \leq x \leq b$$

is continuous on $[a, b]$ and differentiable on (a, b) , and $g'(x) = f(x)$.

Extending Theorem 2.1 with the Chain Rule, we have the following property: Suppose f is continuous on $[a, b]$. If $v(x)$ is a differentiable function whose values lie within $[a, b]$ then,

If

$$g(x) = \int_a^{v(x)} f(t)dt$$

we have

$$g'(x) = \frac{d}{dx} \left(\int_a^{v(x)} f(t)dt \right) = f(v(x)) \frac{d}{dx}(v(x))$$

Given our formulation, we expand the error function into its integral form:

$$f(x) = \frac{x}{2} \left[1 + \operatorname{erf} \left(\frac{\lambda x}{\sqrt{2}} \right) \right] = \frac{x}{2} + \frac{x}{\sqrt{\pi}} \int_0^{\lambda x/\sqrt{2}} e^{-u^2} du$$

Observe that e^{-u^2} is continuous on $\mathbb{R}$, and $\frac{\lambda x}{\sqrt{2}}$ is differentiable on $\mathbb{R}$. By direct application of the product

rule and Theorem 2.1, we compute the derivative:

$$\begin{aligned}
f'(x) &= \frac{d}{dx} \left(\frac{x}{2} + \frac{x}{\sqrt{\pi}} \int_0^{\lambda x/\sqrt{2}} e^{-u^2} du \right) \\
&= \frac{d}{dx} \left(\frac{x}{2} \right) + \frac{d}{dx} \left(\frac{x}{\sqrt{\pi}} \int_0^{\lambda x/\sqrt{2}} e^{-u^2} du \right) \\
&= \frac{1}{2} + \frac{1}{\sqrt{\pi}} \int_0^{\lambda x/\sqrt{2}} e^{-u^2} du + \frac{x}{\sqrt{\pi}} \frac{d}{dx} \int_0^{\lambda x/\sqrt{2}} e^{-u^2} du \\
&= \frac{1}{2} + \frac{1}{\sqrt{\pi}} \int_0^{\lambda x/\sqrt{2}} e^{-u^2} du + \frac{x}{\sqrt{\pi}} e^{-(\lambda x)^2/2} \left(\frac{\lambda}{\sqrt{2}} \right)
\end{aligned}$$

which shows that the proposed activation function, $f(x)$, is continuously differentiable in $\mathbb{R}$.

2.3 Asymptotic Behavior

To demonstrate that the proposed activation function retains the core signal-preserving and noise-filtering properties of standard rectified units, we analyze its asymptotic behavior at the extremes.

First, we examine the behavior for large positive inputs. Given that our scaling parameter is strictly positive ($\lambda \geq 1$), as $x \rightarrow \infty$, the argument of the error function approaches ∞ . By definition, the limit of the error function as its argument approaches ∞ is 1. Evaluating the limit of our formulation:

$$\begin{aligned}
\lim_{x \rightarrow \infty} f(x) &= \lim_{x \rightarrow \infty} \left(\frac{x}{2} \left[1 + \operatorname{erf} \left(\frac{\lambda x}{\sqrt{2}} \right) \right] \right) \\
&= \left(\lim_{x \rightarrow \infty} \frac{x}{2} \right) \cdot \left(\lim_{x \rightarrow \infty} \left[1 + \frac{2}{\sqrt{\pi}} \int_0^{\lambda x/\sqrt{2}} e^{-u^2} du \right] \right) \\
&= \infty \cdot \left[1 + \frac{2}{\sqrt{\pi}} \int_0^{\infty} e^{-u^2} du \right] \\
&= \infty \cdot \left[1 + \frac{2}{\sqrt{\pi}} \left(\frac{\sqrt{\pi}}{2} \right) \right] \quad \left(\text{since } \int_0^{\infty} e^{-u^2} du = \frac{\sqrt{\pi}}{2} \right) \\
&= \infty \cdot (1 + 1) \\
&= \infty
\end{aligned}$$

Although the limit evaluates to ∞ , the function is asymptotically equivalent to the identity function, denoted as $f(x) \sim x$ as $x \rightarrow \infty$. This behavior is critical for deep learning; the necessity of such non-saturating behavior was famously established by Glorot et al. [1], who demonstrated that avoiding the bounded limits of saturating functions (such as Sigmoid or Tanh) on the positive domain is essential for propagating gradients effectively without vanishing. By ensuring $f(x) \sim x$, our proposed activation preserves the identity mapping for active neurons, thereby facilitating stable training in deep architectures.

Next, we examine the behavior for large negative inputs. As $x \rightarrow -\infty$, given our scaling parameter $\lambda \geq 1$, the argument of the error function approaches $-\infty$. Evaluating the limit directly produces an indeterminate form:

$$\lim_{x \rightarrow -\infty} \frac{x}{2} \left[1 + \operatorname{erf} \left(\frac{\lambda x}{\sqrt{2}} \right) \right] \rightarrow -\infty \cdot 0$$

To resolve this, we rewrite the expression as a ratio and apply L'Hôpital's Rule (Theorem 2.2).

Theorem 2.2 (L'Hôpital's Rule). *Let f and g be functions differentiable on an open interval I containing a except possibly at a and $g'(x) \neq 0$ for all $x \in I \setminus \{a\}$.*

If $\lim_{x \rightarrow a} \frac{f(x)}{g(x)}$ is indeterminate of type $\frac{0}{0}$ or $\frac{\infty}{\infty}$, then

$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \lim_{x \rightarrow a} \frac{f'(x)}{g'(x)},$$

provided $\lim_{x \rightarrow a} \frac{f'(x)}{g'(x)}$ exists, or is $+\infty$ or $-\infty$.

By Theorem 2.1

$$\frac{d}{dz} \text{erf}(z) = \frac{d}{dz} \left(\frac{2}{\sqrt{\pi}} \int_0^z e^{-t^2} dt \right) = \frac{2}{\sqrt{\pi}} e^{-z^2}$$

It follows that,

$$\begin{aligned} \lim_{x \rightarrow -\infty} f(x) &= \lim_{x \rightarrow -\infty} \frac{1 + \text{erf}\left(\frac{\lambda x}{\sqrt{2}}\right)}{2x^{-1}} \\ &\stackrel{\text{L'HR}}{=} \lim_{x \rightarrow -\infty} \frac{\frac{2}{\sqrt{\pi}} e^{-\frac{\lambda^2 x^2}{2}} \cdot \frac{\lambda}{\sqrt{2}}}{-2x^{-2}} \\ &= \lim_{x \rightarrow -\infty} \frac{\frac{2}{\sqrt{2\pi}} \lambda e^{-\frac{\lambda^2 x^2}{2}}}{-\frac{2}{x^2}} \\ &= \lim_{x \rightarrow -\infty} -\frac{\lambda}{\sqrt{2\pi}} \left(\frac{x^2}{e^{\frac{\lambda^2 x^2}{2}}} \right) \end{aligned}$$

As $x \rightarrow -\infty$, both x^2 and $e^{\frac{\lambda^2 x^2}{2}}$ approach ∞ , yielding a $\frac{\infty}{\infty}$ indeterminate form. Applying L'Hôpital's Rule again:

$$\begin{aligned} \lim_{x \rightarrow -\infty} f(x) &\stackrel{\text{L'HR}}{=} \lim_{x \rightarrow -\infty} -\frac{\lambda}{\sqrt{2\pi}} \left(\frac{2x}{\lambda^2 x e^{\frac{\lambda^2 x^2}{2}}} \right) \\ &= \lim_{x \rightarrow -\infty} -\frac{\lambda}{\sqrt{2\pi}} \left(\frac{2}{\lambda^2 e^{\frac{\lambda^2 x^2}{2}}} \right) \\ &= 0 \end{aligned}$$

Architecturally, the limit $\lim_{x \rightarrow -\infty} f(x) = 0$ acts as a soft-gating mechanism that induces network sparsity, similar to standard dropout. By suppressing negative inputs, it allows a Multi-Layer Perceptron (MLP) to efficiently isolate complex geometric features without activating the entire parameter space. Furthermore, because this collapse is asymptotic rather than instantaneous, it preserves a small gradient for negative values, mitigating the dead-neuron problem during optimization [1, 2].

Together, these limits mathematically guarantee that the parameterized activation effectively filters out significant negative noise while preserving strong positive signals, maintaining the fundamental advantages of ReLU while providing a continuously differentiable, smooth transition space around the origin.

2.4 Non-Monotonicity and Critical Points

Unlike standard ReLU, which is strictly monotonic, our proposed activation function is non-monotonic, exhibiting a slight negative dip before rising asymptotically. To locate this global minimum, we set the first derivative to zero:

$$f'(x) = \Phi(\lambda x) + \lambda x \phi(\lambda x) = 0$$

Solving this transcendental equation yields a unique negative critical point $x^* < 0$. This localized "negative pocket" allows the network to occasionally assign small negative values to inputs below a certain threshold, providing a soft, adaptive gating mechanism that helps regularize weights and prevent over-saturation during training.

References

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